Def: Let p be an odd prime ($p > 2$)
n is a quadratic residue modulo p
iff $\exists\ x \in \mathbb{Z}_p^\times$ s.t. $x^2 \equiv n \pmod p$

Leg. symbol

$$(n \mid p) = \begin{cases} +1 & \text{if } n \text{ is a residue} \\ -1 & \text{if } n \text{ is a non-residue.} \end{cases}$$

General question: $(n \mid p) = ?$

For any such p, there are $\frac{p-1}{2}$ residues and $\frac{p-1}{2}$ non-residues.

Recall a potentially relevant result
For $n \in \mathbb{Z}_p^\times = U(p)$ Fermat's Little Theorem says

$$n^{p-1} \equiv 1 \pmod p$$

$$\iff \left(n^{\frac{p-1}{2}}\right)^2 \equiv 1 \pmod p$$

This is saying that $n^{\frac{p-1}{2}}$ is a solution to $x^2 \equiv 1 \pmod p$.
which are $x \equiv -1, 1 \pmod p$

Therefore $n^{\frac{p-1}{2}} \equiv -1$ or $1 \pmod{p}$

Is there a connection between $(n|p)$ and $n^{\frac{p-1}{2}}$? YES.

Suppose $(n|p) = 1$, i.e. $n \equiv m^2 \pmod{p}$

$$n^{\frac{p-1}{2}} \equiv (m^2)^{\frac{p-1}{2}} \pmod{p}$$
$$\equiv m^{p-1} \pmod{p}$$
$$\equiv 1 \pmod{p}$$

So $(n|p) = +1 \implies n^{\frac{p-1}{2}} \equiv +1 \pmod{p}$

And so every quad residue n, is a solution to $\underline{x^{\frac{p-1}{2}} \equiv +1} \pmod{p}$

So there are <u>no more solutions to this</u>

So if $(n|p) = -1$, i.e. n is a non-residue.

$$n^{\frac{p-1}{2}} \equiv -1 \pmod{p}$$

This gives us Theorem 8.4 Euler's Criterion

$$(n|p) \equiv n^{\frac{p-1}{2}} \pmod{p}.$$

<u>Application this.</u> $p \equiv 1, 3 \pmod 4$

Let's settle the question

$(-1|p) = ?$

$$(-1|p) \equiv (-1)^{\frac{p-1}{2}} \pmod{p}.$$

$$= (-1)^{\frac{p-1}{2}}$$

$$= \begin{cases} +1 & \text{when } \frac{p-1}{2} \text{ is even} \\ -1 & \text{when } \frac{p-1}{2} \text{ is odd} \end{cases}$$

$$(-1|p) = \begin{cases} +1, & p \equiv 1 \bmod 4 \\ -1, & p \equiv 3 \bmod 4 \end{cases}$$

$p = 4n+3$

$p-1 = 4n+2$

$\frac{p-1}{2} = 2n+1$

What about $(2|p) = ?$

Again we'll apply Euler's Criterion

$$(2|p) \equiv 2^{\frac{p-1}{2}} \pmod{p}.$$

Obtain $2^{\frac{p-1}{2}} \pmod{p}$ in two ways

① $2 \cdot 4 \cdot 6 \cdot 8 \cdots\cdots (p-1) \pmod{p}$

$$\equiv 2^{\frac{p-1}{2}} \left(1 \cdot 2 \cdot 3 \cdots\cdots \frac{p-1}{2}\right) \pmod{p}$$

, factoring

$$\equiv 2^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \pmod{p}.$$

② Again ~~at~~ consider all the even congruence classes

Note.

$$\begin{aligned}
p-1 &\equiv (-1)^1 1 \pmod{p.} \\
2 &\equiv (-1)^2 2 \\
p-3 &\equiv (-1)^3 3 \\
4 &\equiv (-1)^4 4 \\
p-5 &\equiv (-1)^5 5 \\
&\vdots \\
&\equiv (-1)^{\frac{p-1}{2}} \frac{p-1}{2}
\end{aligned}$$

Now multiply all these together.

$$\prod_{i=1}^{p-1/2} 2i \equiv \prod_{i=1}^{(p-1)/2} (-1)^i i$$

$$\equiv \left[\prod_{i=1}^{(p-1)/2} (-1)^i\right]\left[\prod_{i=1}^{(p-1)/2} i\right]$$

$$\equiv (-1)^{\sum_{i=1}^{(p-1)/2} i} \left(\frac{p-1}{2}\right)!$$

$$\equiv (-1)^{\frac{p-1}{4}\frac{p+1}{2}} \left(\frac{p-1}{2}\right)!$$

$$a^b a^c = a^{b+c}$$

$$\equiv (-1)^{(p^2-1)/8} \left(\frac{p-1}{2}\right)!$$

$$\sum_{j=1}^{m} j = \frac{m}{2}(m+1)$$

Equating ① $\equiv$ ② $\pmod p$

$$\Rightarrow 2^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \equiv (-1)^{(p^2-1)/8} \left(\frac{p-1}{2}\right)!$$

$$\Rightarrow 2^{\frac{p-1}{2}} \equiv (-1)^{(p^2-1)/8}$$

$$\equiv \begin{cases} +1, & (p^2-1)/8 \text{ is even} \\ -1, & (p^2-1)/8 \text{ is odd.} \end{cases}$$

$$p \equiv 1, 3, 5, 7 \pmod 8$$

(see Table 8.1 in notes)

$$(2|p) \equiv \begin{cases} +1 & , p \equiv 1, 7 \pmod 8 \\ -1 & , p \equiv 3, 5 \pmod 8. \end{cases}$$

Could continue in this way.
Analyzing $(3|p)$, $(5|p)$ etc.
But instead, recall our analysis.

$$n = \prod_{i=1}^{r} p_i^{a_i} \quad , \qquad p \neq p_i \quad i = 1, \dots, r$$

$$(n|p) = \left(\prod p_i^{a_i} \middle| p \right)$$

$$= \prod (p_i|p)^{a_i}$$

$$= \prod_{\substack{i=1 \\ a_i \text{ odd}}}^{r} (p_i|p).$$

So let's examine $(p|q)$.
for a pair of distinct odd primes p, q.

Continue at 4:55

From plot

$\forall$ prime $p \neq 197$ $(p \mid 197) = (197 \mid p)$

Sometimes a row matches the reflected column.

The comparison reveals an alternating pattern of "all white" / "mixed" columns going over the odd integers, ie. a pattern modulo 4.

Also, going up any "mixed" column we see an alternating pattern of black/white across all the odd integers ie. a pattern modulo 4.

So this comparison reveals the Law of Quadratic Reciprocity.

For odd distinct primes p, q

$$(p \mid q) = \begin{cases} -(q \mid p), & p \equiv q \equiv 3 \pmod 4 \\ (q \mid p), & \text{otherwise.} \end{cases}$$

$p \equiv 1, 3 \pmod 4$

eg. $(3|7) = -(7|3)$, since $7 \equiv 3 \bmod 4$

in fact $(3|7) = -1$

and $(7|3) = (1|3) = 1$

But $(5|7) = (7|5)$, since $5 \equiv 1 \bmod 4$

in fact $(5|7) = -1$

$(7|5) = (2|5) = -1$

We won't be examing the proof of LQR.

One way of proving it is presented in notes.

But we will study the following algorithm for determing $(n|p)$ based on a prime factorisation of n.

LQR

$(p|q) = \pm(q|p)$

FLIP

$(ab|p)$

$= (a|p)(b|p)$ FACTORIZE

REDUCE

$n \equiv m \mod p$

$(n|p) = (m|p)$

Eg. Compute.

$(219|383)$, 383 prime.

$= (73 \cdot 3 | 383)$

$= (73|383)(3|383)$, Factorizing

$= (383|73)(-(383|3))$, LQR. $73 \equiv 1 \pmod 4$, $383 \equiv 3 \pmod 4$

$= (18|73)(-(2|3))$, reducing mod 73 & mod 3

$= (18|73)$, since $(2|3) = -1$

$= (2 \cdot 3^2 | 73)$

$= (2|73)(3|73)^2$, Factorizing.

$= (2|73)$, since $(3|73) = \pm 1$

$= +1$, since $73 \equiv 1 \mod 8$ and by the theorem for $(2|p)$

Exercises Q 11

Compute. Both 461, 773 are prime.

$(461 \mid 773)$

$= (773 \mid 461)$, by L.Q.R. and $461 \equiv 1 \pmod 4$

$= (312 \mid 461)$, by congruence prop. REDUCE $773 \equiv 312 \mod 461$

$= (2^3 \cdot 3 \cdot 13 \mid 461)$

$312 = 3 \cdot 104$

$= (2 \mid 461)^3 (3 \mid 461)(13 \mid 461)$

$= 2 \cdot 3 \cdot 52$

$= 4 \cdot 3 \cdot 26$

$= 8 \cdot 3 \cdot 13$

Factorizing.

$= (2 \mid 461)(3 \mid 461)(13 \mid 461)$

$= -(3 \mid 461)(13 \mid 461)$, using $(2 \mid p)$ and $461 \equiv 5 \mod 8$

$= -(461 \mid 3)(461 \mid 13)$, L.Q.R. $461 \equiv 1 \mod 4$

$= -(2 \mid 3)(6 \mid 13)$

$= (6 \mid 13)$, since $(2 \mid 3) = -1$

$= (2|13)(3|13)$, Factorizing

$= -(3|13)$, $(2|p)$ result $13 \equiv 5 \pmod 8$

$= -(13|3)$, by LQR, $13 \equiv 1 \bmod 4$

$= -(1|3)$

$= -1$, since $(1|3) = +1$